ELEC-E8126 - Robotic Manipulation

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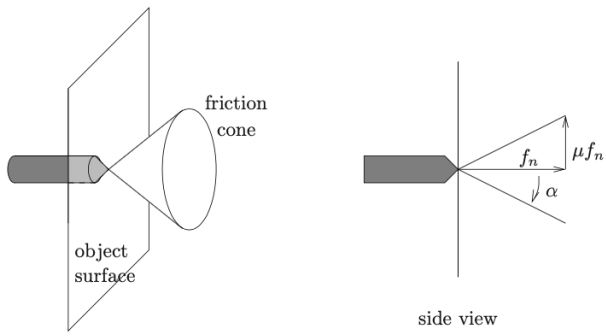
Exercise 5 + (Optional)

Estimated Time (Code: 60 mins, Report: 60 mins, Study: 240 mins)

Considering on the edges of our target polygon, each infinitesimal point could be grasp contact, so it makes sense to first sampling on all the sides of polygon as candidates.

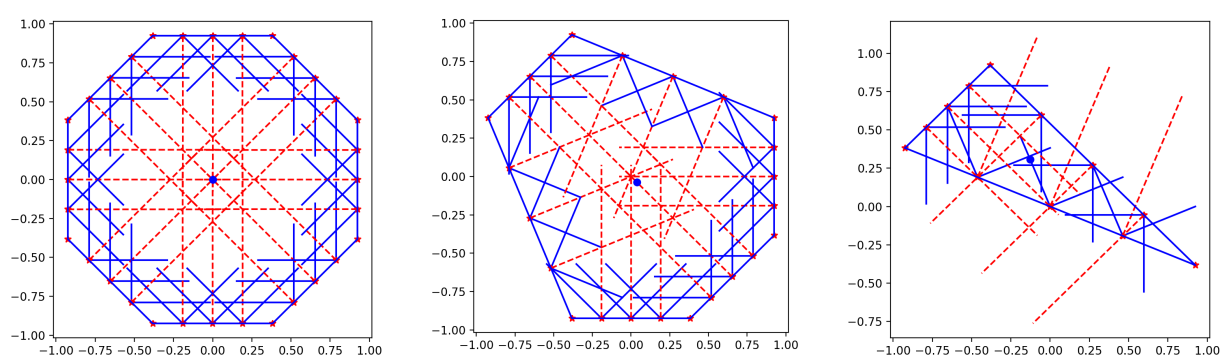
All the possible grasp contacts only a few pair could actual make object stable, and for this assignment assumed that *friction coefficient* is **1** for all kind of geometry shapes (objects). Then according to *Coulomb friction model*, we know that contact force must line in the following region called *friction cone*, and the angle here with respect to the normal is equal $\pm 45^\circ$

$$\left|f^t\right| \leq \mu f^n \text{ and } \alpha = tan^{-1}\mu = 45^\circ$$



In this assignment, we also assumed that grasp contact approach direction must perpendicular to object (2D) surface, so for later calculation is more simple.

```
SAMPLING_NUM = 5
```



To execute grasp planning code and review the given result.

You could simply type the following command to terminal.

Code

```
python3 run.py
```

Graph

- grasping points (sampling points)
- approach direction (perpendicular line)
- friction cones region (+-45)

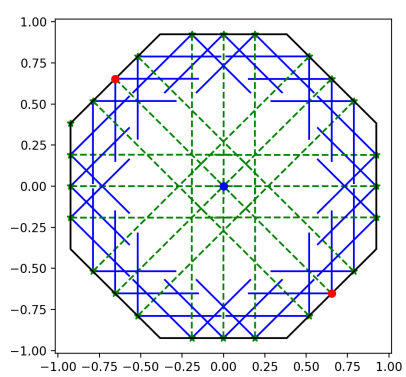
Graph Data

data/test.txt **-> all the possible candidates**

data/grasp.txt **-> stable grasp**

The concept of stable grasp contacts (red dots):

If pair grasp contacts (line) lies in the friction cones then grasp is stable.



Library:

Shapely, Matplotlib, numpy, math

– Is the grasp both force-closure and form-closure, either of, or neither? Explain why.

According to "A Mathematical Introduction to Robotic Manipulation", Theorem 5.6. Planar antipodal grasps [82] , we know that two soft-finger contacts in this assignment is a force-closure because the contact point lies inside both friction cones.

And if a rigid body is fully immobilized by a set of rigid stationary fixtures we called it form-closure, however, in this assignment a planar body at least need 4 point contacts, so it is not a form-closure.

– If the object was not force-closure, where would we need to place additional contacts to achieve it?

The question is if there is no any pair of grasping contact point to form a force-closure, then where to put another grasping point to achieve force closure.

By the definition that a set of friction grasp contacts forms a force closure only if the positive span of the wrench cones is equal to the entire wrench space.

$F = [F_1 \dots F_j] \in R^{n \times j}$

Force Closure iff:

- $rank(F) = n$ (3 or 6)
- $Fk = 0$ for some $k \in R^j, k \geq \epsilon > 0$ for all i

Reference:

Constructing Force-Closure Grasps
https://journals.sagepub.com/doi/abs/10.1177/027836498800700301

Optional (folder name - dex-net)

The workflow is similar in the assignment but extending to 3D object using Dex-Net by UC Berkeley. (https://berkeleyautomation.github.io/dex-net/)

(1) Sampling Method Dex-Net provides in total five sampling methods (UniformGraspSampler, GaussianGraspSampler, AntipodalGraspSampler, GpgGraspSampler and PointGraspSampler)

Our assignment only considering two-finger robotic gripper, so we will only look into **AntipodalGraspSampler** method

(2) Grasp Contact Model About contact point and grasp is well-defined in Dex-Net 1.0 paper which assumed robotic gripper contact using *Coulomb friction model*, and the class **ParallelJawPtGrasp3D** define robotic grasp in 3D space that each grasp has two contact points (two-finger gripper).

From those two contact points will compute the **friction cone** and **normal** and then **return a list of candidate grasps for graspable object** (in the class GraspableObject3D)

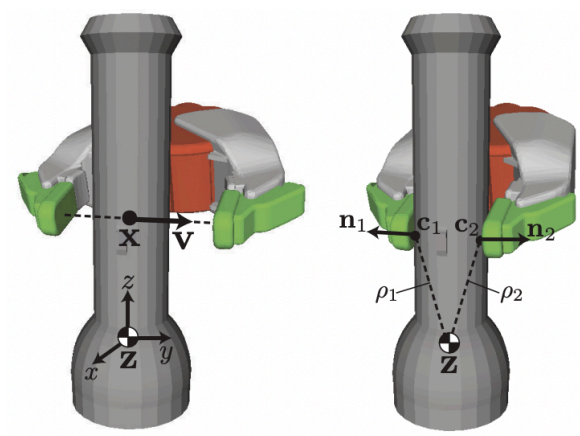
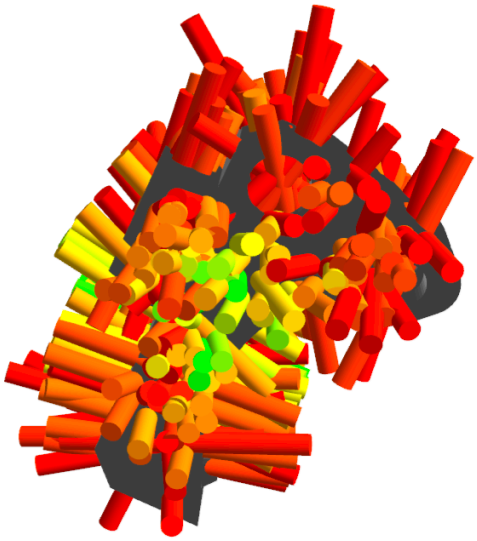
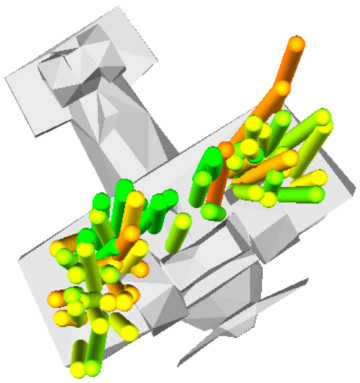


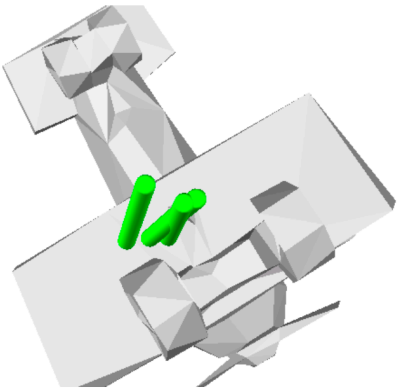
Illustration of grasp parameterization and contact model - more details in **Dex-Net 1.0**

Quality Metrics When sampling method return candidate grasps, we will use quality metrics and set parameters to filter out interested grasps, later on checking both Ferrari-Canny Grasps Robustness Metric and Force Closure to generate reasonable grasp result.





(a) Robotic Grasp Robustness: 10



(b) Robotic Grasp Robustness: 90

YCB Object Dataset - 072-a_toy_airplane (google_512k)